

NETFLIX CONTENT LIBRARY

Data Analysis Capstone Project

Complete Findings Report

PROJECT OVERVIEW

Three datasets | 8,800 deduplicated titles | 5 business goals | 15 SMART questions

Tools: Excel | Google Sheets | BigQuery (SQL) | Tableau Public

Prepared by: Geri Rusk | Date: June 11, 2026

1. Project Background

This project was undertaken as a beginner data analyst portfolio piece and capstone project, covering the full data analysis lifecycle from raw data acquisition through cleaning, SQL analysis, and data visualization. The analysis examines Netflix's content library to uncover strategic insights about content quality, genre performance, global reach, and trends over time.

1.1 Datasets Used

Dataset	Source	Rows	Key Columns
netflix_titles.csv	Kaggle	8,807	title, type, country, date_added, rating, listed_in
Netflix_TV_Shows_and_Movies.csv	Kaggle	5,283	title, imdb_score, imdb_votes, runtime
netflix-rotten-tomatoes-metacritic-imdb.csv	Kaggle	15,480	title, rt_score, hidden_gem_score, genre

1.2 Data Quality Summary

The following data quality issues were identified and resolved prior to analysis:

- 159 duplicate title records identified and resolved using ROW_NUMBER() window function in BigQuery, retaining the record with highest IMDB vote count for each duplicate group
- 10 titles had no recorded date_added value – cells left blank for NULL handling in BigQuery and excluded from time-based analyses only
- Director column had 30% missing values – filled with Unknown placeholder
- Country column had 9% missing values - filled with Unknown placeholder
- Age certification column in netflix_scores (Netflix_TV_Shows_and_Movies) had 43% missing values – primary rating column from netflix_titles used instead
- Rotten Tomatoes scores missing for 59% of titles in netflix_rt (netflix-rotten-tomatoes-metacritic-imdb) used where available with coverage noted
- Non-Latin character titles (Arabic, Sanskrit) cause BigQuery encoding issues – resolved by saving all CSV files in UTF-8 format
- Two “Death Note” records retained as distincy entries representing a 2006 Japanese anime series and a 2017 US live action film

1.3 Master Dataset

A deduplicated master view (netflix_master_deduped) was created in BigQuery joining all three datasets on the title column using LEFT JOIN to preserve all 8,800 titled regardless of score availability.

Metric	Count
Total unique titles (after deduplication)	8,800
Titles with IMDB scores	4,163 (47.3%)
Titles with Rotten Tomatoes scores	2,268 (25.8%)
Titles missing IMDB scores	4,637 (52.7%)

2. Business Goals & SMART Questions

All analysis in this project was structured around five business goals, each supported by two to three SMART questions. SMART questions are Specific, Measurable, Action-oriented, Relevant, and Time-bound. Each question was designed to be answerable using the available dataset columns and SQL queries.

Goal 1 – Content Quality Assessment

Understand how Netflix content performs in terms of audience and critic perception, and identify which content types and genres consistently deliver high-quality ratings.

Q1.1 – *What is the average and median IMDB score for Movies versus TV Shows on Netflix, and which content type has a higher percentage of titles scoring above 7.0?*

Q1.2 – *Which 10 titles have the highest IMDB scores and which have the lowest, and what do they have in common in terms of type, genre, or country?*

Q1.3 – *Is there a relationship between the number of IMDB votes a title receives and its IMDB score, i.e. do more widely-watched titles tend to rate higher?*

Goal 2 – Content Strategy by Genre

Identify which genres are over- or under-represented relative to their IMDB performance, to inform future content investment decisions.

Q2.1 – *What are the 5 most-produced genres on Netflix, and what is the average IMDB score for each, i.e. are the most-produced genres also the highest-rated?*

Q2.2 – *Which genres have the highest average IMDB score but the fewest titles representing potential underinvested opportunities for Netflix?*

Goal 3 – Global Content Reach

Evaluate the geographic diversity of Netflix's content library and determine whether international content performs differently than US-produced content.

Q3.1 – Which 10 countries produce the most Netflix content, and how does their average IMDB score compare to US-produced content?

Q3.2 – What percentage of Netflix's library is non-US content, and has that percentage grown or shrunk between 2016 and 2021?

Q3.3 – Which country outside the United States produces the highest average IMDB score, with at least 20 titles to ensure statistical reliability?

Goal 4 – Critic vs Audience Alignment

Determine whether content maturity ratings correlate with IMDB scores, and whether critic scores and audience scores tell the same story.

Q4.1 – Is there a meaningful difference in average IMDB score between TV-MA rated content and family-friendly content (G, PG, TV-G, TV-Y), and which rating category performs best?

Q4.2 – For titles where both IMDB score and a Rotten Tomatoes score exist, how often do they agree with this being defined as both being above or below their respective averages?

Q4.3 – Which genres show the biggest gap between critic scores (Rotten Tomatoes) and audience scores (IMDB), and in which direction does the gap go?

Goal 5 – Library Growth & Quality Trends Over Time

Analyze how Netflix's content volume and average quality ratings have changed year over year, to identify whether growth periods impacted content quality.

Q5.1 – How many titles were added to Netflix each year between 2015 and 2021, broken down by Movies and TV Shows, and which year had the largest single-year increase?

Q5.2 – *Did the average IMDB score of Netflix content added each year increase or decrease between 2015 and 2021; and if it changed, by how much?*

Q5.3 – *What is the average gap between a title's original release year and the year it was added to Netflix, and has that gap changed over time, suggesting Netflix is licensing older or newer content?*

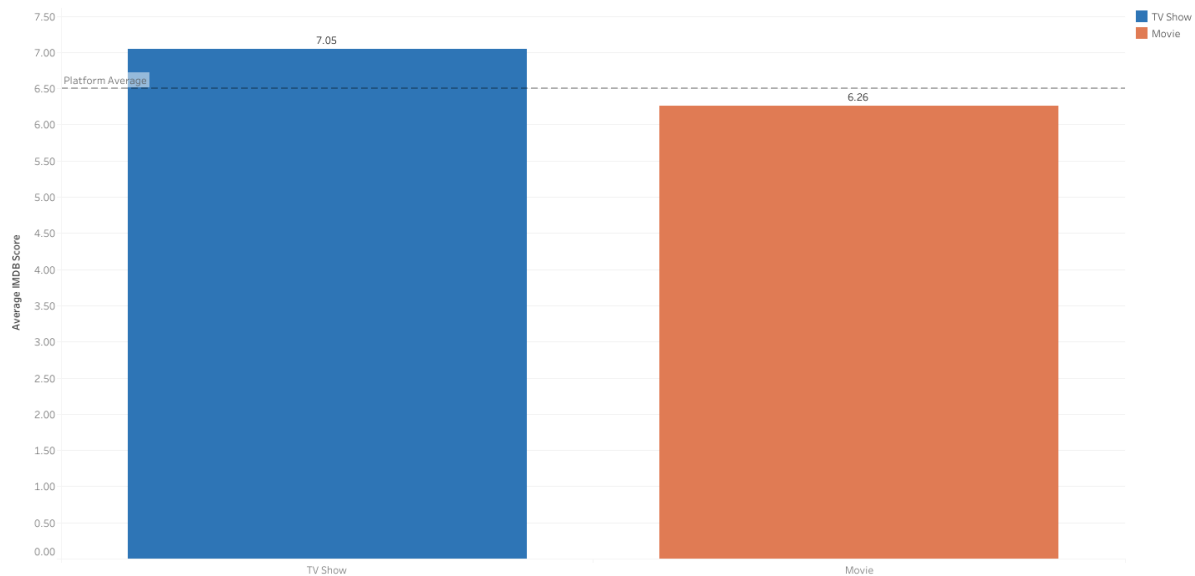
GOAL 1: CONTENT QUALITY ASSESSMENT

Charts 1 & 2 | Q1.1, Q1.2, Q1.3 | Dashboard 1

Q1.1 – Average IMDB Score by Content Type

Q1.1 – What is the average and median IMDB score for Movies versus TV Shows on Netflix, and which content type has a higher percentage of titles scoring above 7.0?

TV Shows vs Movies - Average IMDB Score



Content Type	Avg IMDB Score	% Scoring Above 7.0	Titles with Scores
TV Show	7.05	55.5%	1,476
Movie	6.26	24.7%	2,439

Key Insight: TV Shows on Netflix consistently outperform Movies in audience perception, averaging 7.05 vs 6.26 on IMDB, a difference of 0.8 points. TV Shows are more than twice as likely to score above 7.0 (55.5% vs 24.7%). This finding remained consistent before and after duplicate removal, confirming its reliability.

Q1.2 – Top 10 and Bottom 10 Titles by IMDB Score

Q1.2 – Which 10 titles have the highest IMDB scores and which have the lowest, and what do they have in common in terms of type, genre, or country?

Top 10 Highest Rated Titles:

Query results							
Job information		Results	Visualization	JSON	Execution details	Execution graph	
Row	title_cleaned	type_cleaned	country_cleaned	listed_in	imdb_score	imdb_votes	
1	Khawatir	TV Show	Unknown	International TV Shows, Reality ...	9.6	3046	
2	Breaking Bad	TV Show	United States	Crime TV Shows, TV Dramas, T...	9.5	1727694	
3	Kota Factory	TV Show	India	International TV Shows, Roman...	9.3	66985	
4	Our Planet	TV Show	United States, United Kingdom	Docuseries, Science & Nature TV	9.3	41386	
5	Avatar: The Last Airbender	TV Show	United States	Classic & Cult TV, Kids' TV, TV A...	9.3	297336	
6	My Mister	TV Show	South Korea	Crime TV Shows, International T...	9.2	5481	
7	Reply 1988	TV Show	South Korea	International TV Shows, Korean ...	9.2	6557	
8	The Last Dance	TV Show	United States	Docuseries	9.1	108321	
9	Death Note	TV Show	Japan	Anime Series, Crime TV Shows, ...	9.0	302147	
10	Death Note	TV Show	Japan	Anime Series, Crime TV Shows, ...	9.0	302147	

Title	Type	Country	IMDB Score	IMDB Votes
Khawatir	TV Show	Unknown	9.6	3,046
Breaking Bad	TV Show	United States	9.5	1,727,694
Kota Factory	TV Show	India	9.3	66,985
Our Planet	TV Show	US / UK	9.3	41,386
Avatar: The Last Airbender	TV Show	United States	9.3	297,336
My Mister	TV Show	South Korea	9.2	5,481
Reply 1988	TV Show	South Korea	9.2	6,557
The Last Dance	TV Show	United States	9.1	108,321
Death Note	TV Show	Japan	9.0	302,147

Bottom 10 Lowest Rated Titles:

Query results							
Job information		Results	Visualization	JSON	Execution details	Execution graph	
Row	title_cleaned	type_cleaned	country_cleaned	listed_in	imdb_score	imdb_votes	
1	Aerials	Movie	United Arab Emirates	International Movies, Sci-Fi & Fa...	1.5	382	
2	Himmatwala	Movie	India	Action & Adventure, Comedies, I...	1.7	8550	
3	Racket Boys	TV Show	Unknown	International TV Shows, TV Co...	1.8	27484	
4	Kyaa Kool Hain Hum 3	Movie	India	Comedies, International Movies	1.9	4341	
5	Hajwala: The Missing Engine	Movie	United Arab Emirates	Action & Adventure, Dramas, Int...	2.1	250	
6	Dostana	Movie	India, United States	Comedies, Dramas, Internationa...	2.1	25	
7	Time To Dance	Movie	India	Dramas, International Movies, R...	2.2	950	
8	Myriam Fares: The Journey	Movie	United Arab Emirates	Documentaries, International M...	2.3	153	
9	A House Of Blocks	TV Show	Taiwan	International TV Shows, TV Dra...	2.3	137	
10	Romina	Movie	Mexico	Horror Movies, Independent Mo...	2.3	2282	

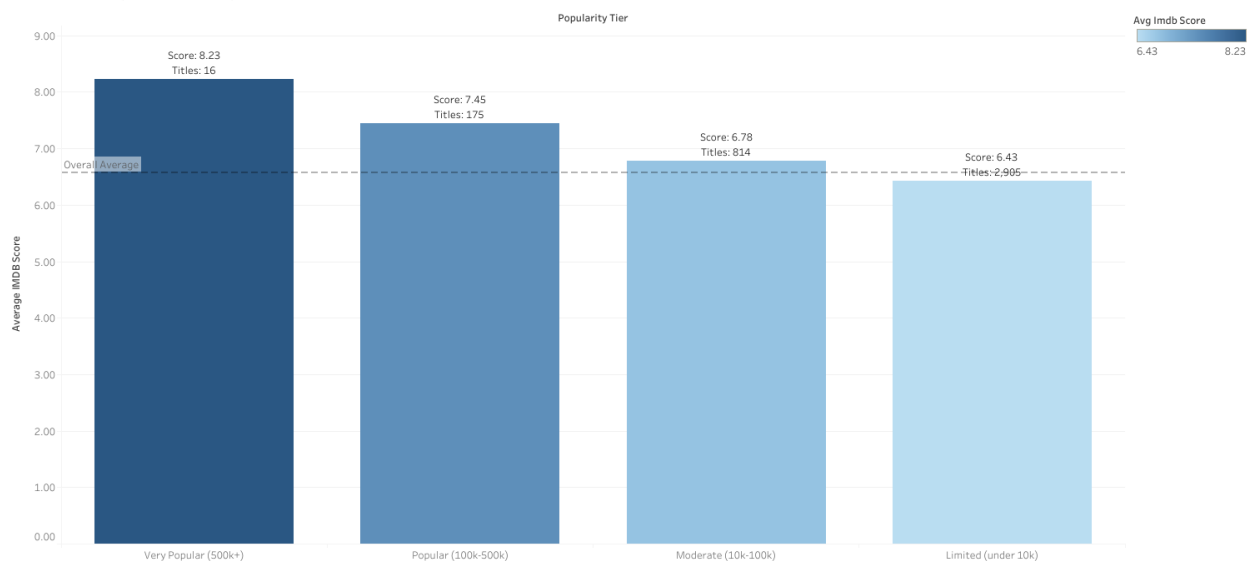
Title	Type	Country	IMDB Score	IMDB Votes
Aerials	Movie	United Arab Emirates	1.5	382
Himmatwala	Movie	India	1.7	8,550
Racket Boys	TV Show	Unknown	1.8	27,484
Kyaa Kool Hain Hum 3	Movie	India	1.9	4,341
Hajwala: The Missing Engine	Movie	UAE	2.1	250
Dostana	Movie	India / US	2.1	25
Time to Dance	Movie	India	2.2	950
Myriam Fares: The Journey	Movie	UAE	2.3	153
A House of Blocks	TV Show	Taiwan	2.3	137
Romina	Movie	Mexico	2.3	2,282

Key Insight: The top 10 highest rated titles are exclusively TV Shows, with strong international representation from India, South Korea, and Japan. The bottom 10 is dominated by Movies (8 of 10), with content from India and the UAE appearing frequently at both extremes, indicating high quality variance in Netflix's international Movie catalog. Several bottom-ranked titles have very few IMDB votes, raising questions about statistical reliability for low-engagement content.

Q1.3 – Relationship Between IMDB Votes and Score

Q1.3 – Is there a relationship between the number of IMDB votes a title receives and its IMDB score, i.e. do more widely-watched titles tend to rate higher?

Does Popularity Equal Quality?



Popularity Tier	Title Count	% of Total	Avg IMDB Score
Very Popular (500k+ votes)	16	0.4%	8.23
Popular (100k-500k votes)	175	4.5%	7.45
Moderate (10k-100k votes)	814	21.0%	6.78
Limited (under 10k votes)	2,905	74.8%	6.43

Key Insight: A strong positive correlation exists between audience engagement and content quality on Netflix. Titles with over 500,00 IMDB votes average 8.23, nearly 1.8 points higher than titles with fewer than 10,000 votes. However, 74.8% of Netflix's scored catalog falls in the lowest engagement tier, meaning platform averages are heavily influenced by low-visibility content. Netflix's most-watched content significantly outperforms its catalog average, suggesting a quality divide between marquee titles and the broader library.

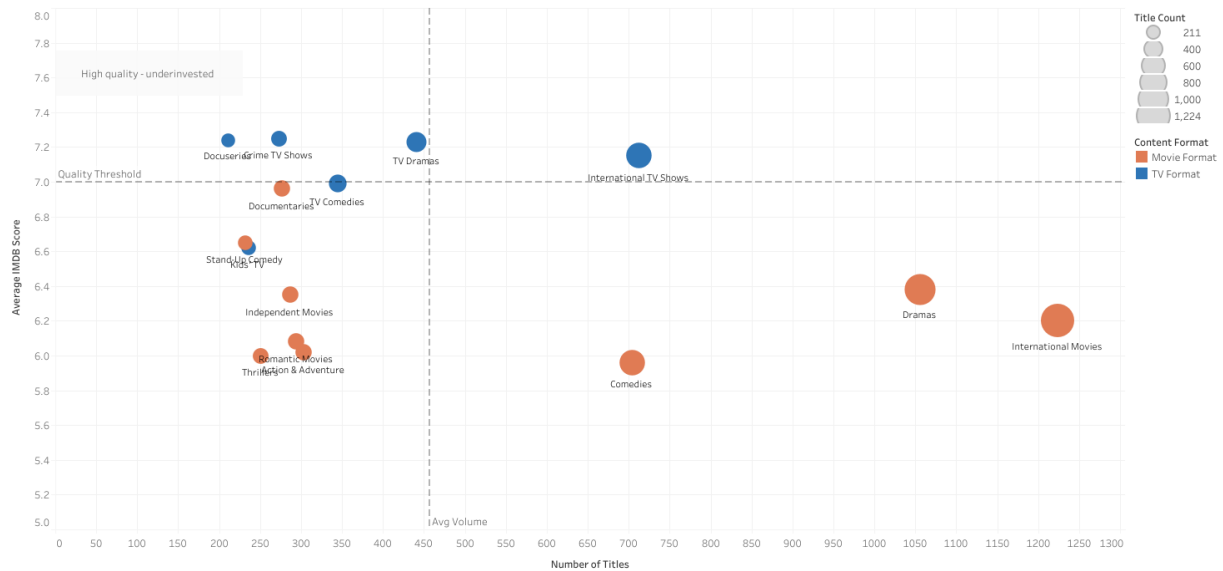
GOAL 2: CONTENT STRATEGY BY GENRE

Charts 3 & 4 | Q2.1, Q2.2, | Dashboard 1

Q2.1 – Most Produced Genres vs Average IMDB Score

Q2.1 – What are the 5 most-produced genres on Netflix, and what is the average IMDB score for each, i.e. are the most-produced genres also the highest-rated?

Are Netflix's Most Produced Genres Its Best?



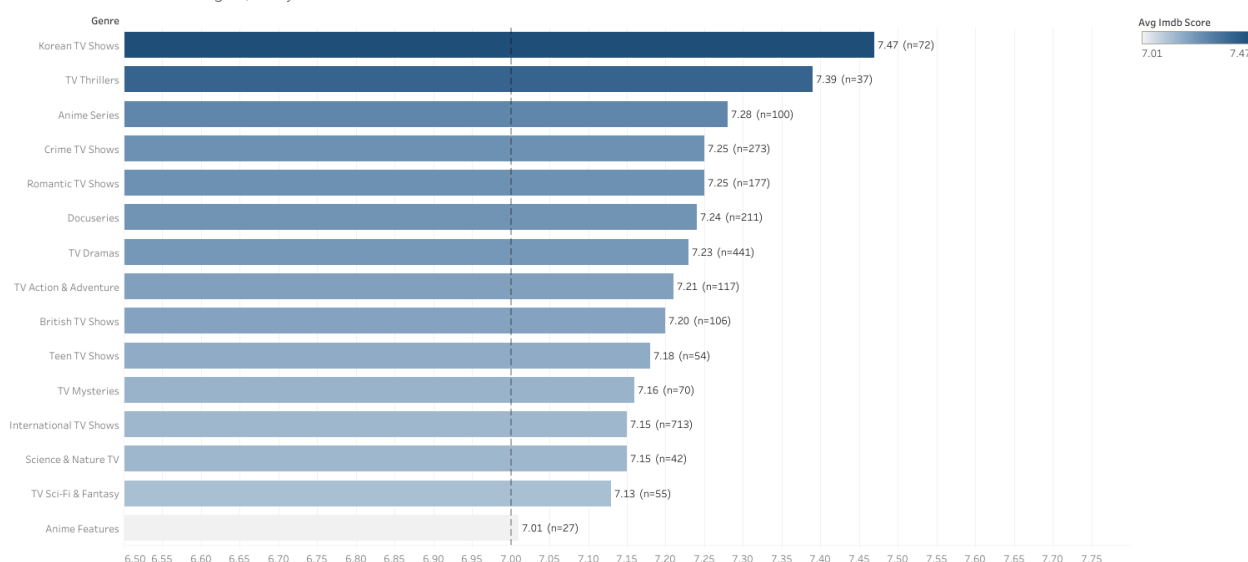
Genre	Title Count	Avg IMDB Score
International Movies	1,224	6.20
Dramas	1,056	6.38
International TV Shows	713	7.15
Comedies	704	5.96
TV Dramas	441	7.23
TV Comedies	345	6.99
Action & Adventure	303	6.02
Romantic Movies	294	6.08
Independent Movies	287	6.35
Documentaries	277	6.96
Crime TV Shows	273	7.25
Thrillers	251	6.00
Kids TV	236	6.62
Stand-Up Comedy	232	6.65
Docuseries	211	7.24

Key Insight: Netflix's most produced genres are not its highest rated ones. International Movies and Dramas dominate the catalog with over 1,000 titles each but average below 6.4 on IMDB. Every TV genre in the top 15 outperforms every Movie genre, this suggests that Netflix's content volume strategy in Movies is not aligned with audience quality expectations. Comedies (704 titles, 5.96 avg) represent a particularly concerning volume-quality misalignment.

Q2.2 – High Rated but Underproduced Genres (min. 20 titles)

Q2.2 – Which genres have the highest average IMDB score but the fewest titles representing potential underinvested opportunities for Netflix?

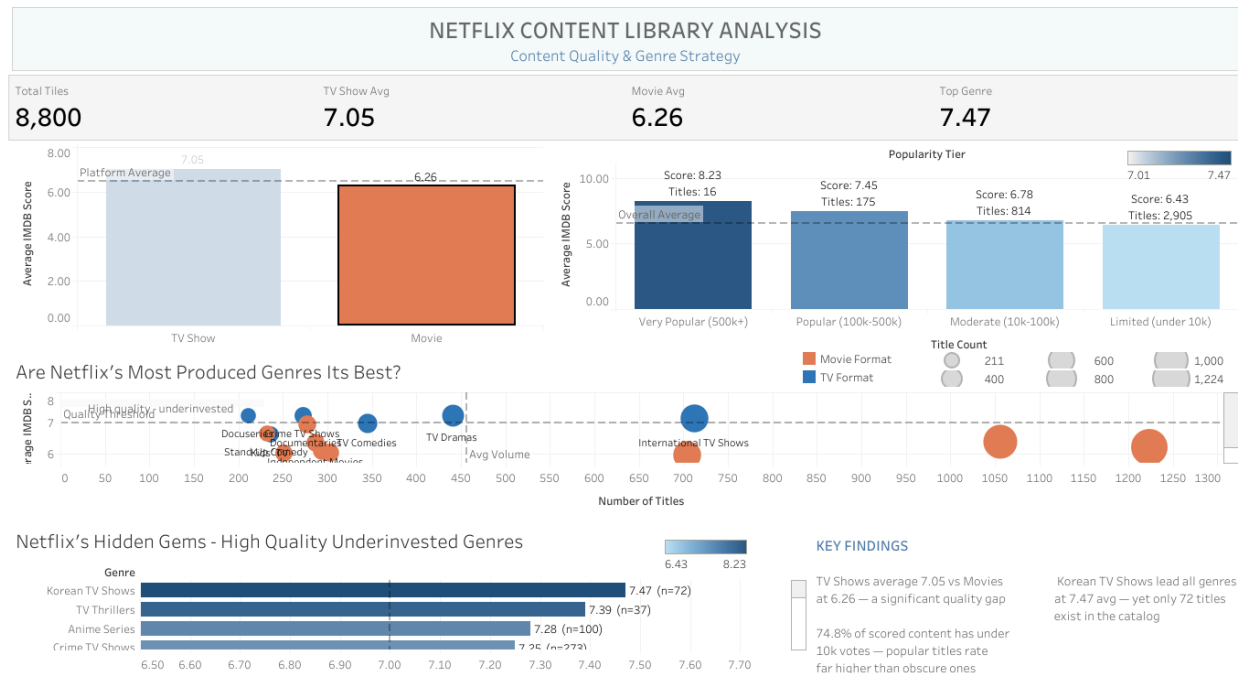
Netflix's Hidden Gems - High Quality Underinvested Genres



Genre	Title Count	Avg IMDB Score	Opportunity Assessment
Korean TV Shows	72	7.47	Highest rated, significantly underinvested
TV Thrillers	37	7.39	High score, very small catalog
Anime Series	100	7.28	Strong score, room to expand
Romantic TV Shows	177	7.25	Well-sized but could grow
Crime TV Shows	273	7.25	High volume and quality, invest more
Docuseries	211	7.24	Strong performer
TV Dramas	441	7.23	Large and strong, priority format
TV Action & Adventure	117	7.21	Underinvested relative to quality
British TV Shows	106	7.20	Quality niche
Science & Nature TV	42	7.15	High score, very small catalog

Key Insight: Korean TV Shows represent Netflix's highest rated genre at 7.47 average IMDB score, yet comprise only 72 titles, showing a significantly underrepresented catalog relative to quality performance. TV Thrillers (7.39), Anime Series (7.28), and Science & Nature TV (7.15) similarly show strong audience reception with limited catalog depth. All 15 highest rated genres are TV formats, reinforcing a consistent platform-wide quality advantage for TV content over Movies.

Dashboard 1: Content Quality & Genre Strategy



KEY FINDINGS:

- TV Shows average 7.05 vs Movies at 6.26, which shows a significant quality gap
- 74.8% of scored content has under 10k votes; popular titles rate far higher than obscure ones
- Korean TV Shows lead all genres at 7.47 avg, yet only 72 titles exist in the catalog

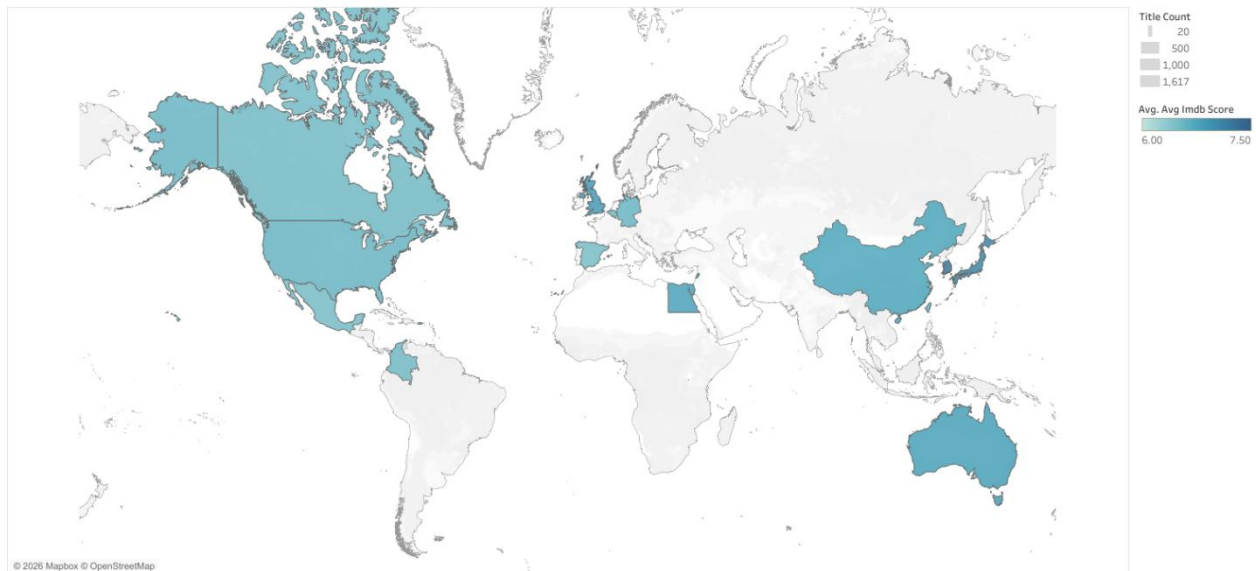
GOAL 3: GLOBAL CONTENT REACH

Charts 5 & 6 | Q3.1, Q3.2, Q3.3 | Dashboard 2

Q3.1 – Top 10 Producing Countries vs IMDB Score

Q3.1 – Which 10 countries produce the most Netflix content, and how does their average IMDB score compare to US-produced content?

Where Does Netflix's Best Content Come From?

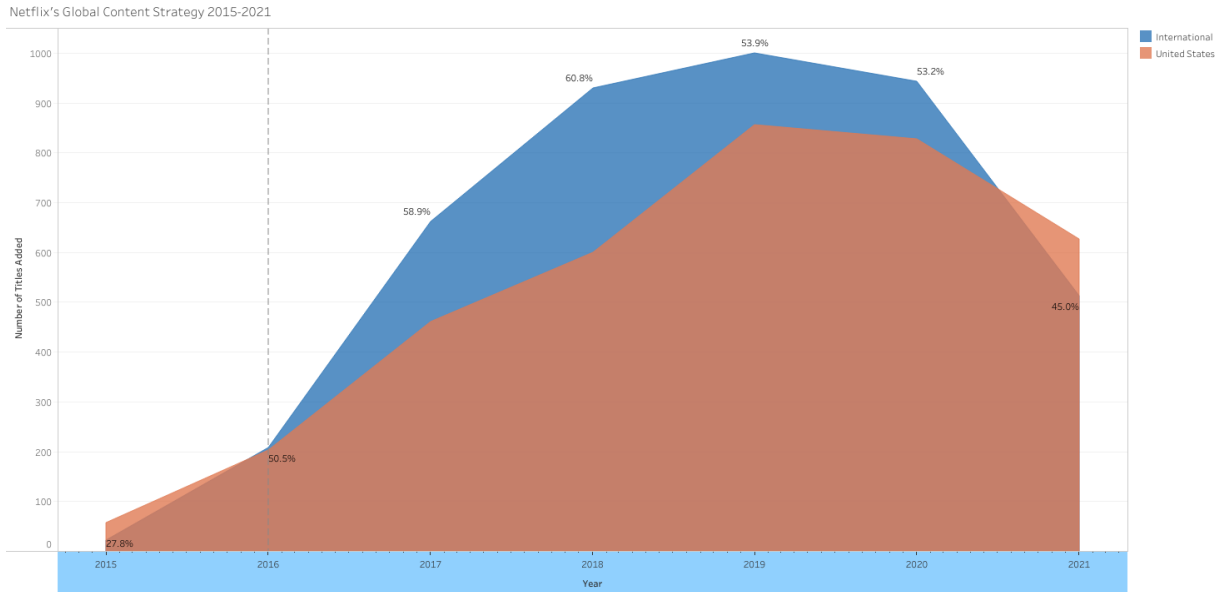


Rank	Country	Title Count	Avg IMDB Score	vs US Score
1	United States	1,617	6.58	Baseline
2	India	485	6.36	-0.22
3	United Kingdom	295	6.87	+0.29
4	France	177	6.45	-0.13
5	Canada	174	6.51	-0.07
6	Japan	166	7.10	+0.52
7	Spain	133	6.48	-0.10
8	South Korea	124	7.25	+0.67
9	Germany	96	6.58	0.00
10	China	83	6.75	+0.17

Key Insight: The United States dominates Netflix's content library with 1,617 scored titles, over three times more than the second largest producer, India. However, volume does not equal quality. South Korea (7.25) and Japan (7.10) significantly outperform the US average of 6.58 despite having far fewer titles. India, despite being the second largest content contributor, ranks near the bottom of quality scores at 6.36.

Q3.2 – International vs US Content Growth Over Time

Q3.2 – What percentage of Netflix’s library is non-US content, and has that percentage grown or shrunk between 2016 and 2021?



Year	Total Titles	International	US Titles	% International
2015	79	22	57	27.8%
2016	410	207	203	50.5%
2017	1,122	661	461	58.9%
2018	1,528	929	599	60.8%
2019	1,856	1,000	856	53.9%
2020	1,771	943	828	53.2%
2021	1,138	512	626	45.0%

Key Insight: Netflix’s globalization strategy is clearly visible in the data. International content grew from just 27.8% of the catalog in 2015 to a peak of 60.8% in 2018, coinciding with Netflix’s 2016 expansion into 130 new countries. Between 2016 and 2020, international titles consistently represented over half of all new content added annually. The 2021 drop to 45% likely reflects a partial year in the dataset rather than a strategic reversal.

Q3.3 – Highest Rated Countries (min. 20 titles)

Q3.3 – Which country outside the United States produces the highest average IMDB score, with at least 20 titles to ensure statistical reliability?

Rank	Country	Title Count	Avg IMDB Score
1	South Korea	124	7.25
2	Japan	166	7.10
3	United Kingdom	295	6.87
4	Lebanon	23	6.86
5	Australia	65	6.82
6	Egypt	25	6.81
7	China	83	6.75
8	Taiwan	38	6.64
9	United States	1,617	6.58
10	Germany	96	6.58

Key Insight: Among all countries with at least 20 titles, South Korea ranks first at 7.25 followed by Japan at 7.10. Notably Lebanon ranks 4th at 6.86 and Egypt 6th at 6.81, both outperforming the United States which ranks 9th despite producing over 1,600 titled. This suggests Netflix has untapped quality opportunities in Middle Eastern content markets that remain significantly underinvested relative to their quality output.

GOAL 4: CRITIV VS AUDIENCE ALIGNMENT

Charts 7 & 8 | Q4.1, Q4.2, Q4.3 | Dashboard 2

Q4.1 – Content Rating vs IMDB Score

Q4.1 – Is there a meaningful difference in average IMDB score between TV-MA rated content and family-friendly content (G, PG, TV-G, TV-Y), and which rating category performs best?

Query results

Chat

View all jobs

Save results

Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	rating_cleaned	title_count	avg_imdb_score	pct_above_7
1	TV-Y7-FV	1	6.8	0.0
2	TV-MA	1814	6.63	38.0
3	TV-14	1003	6.61	41.9
4	TV-PG	316	6.56	34.5
5	TV-Y7	121	6.53	35.5
6	TV-Y	120	6.47	33.3
7	PG-13	126	6.38	24.6
8	G	5	6.38	40.0
9	TV-G	102	6.34	27.5
10	R	228	6.16	21.5
11	NC-17	1	6.1	0.0
12	PG	76	6.08	13.2

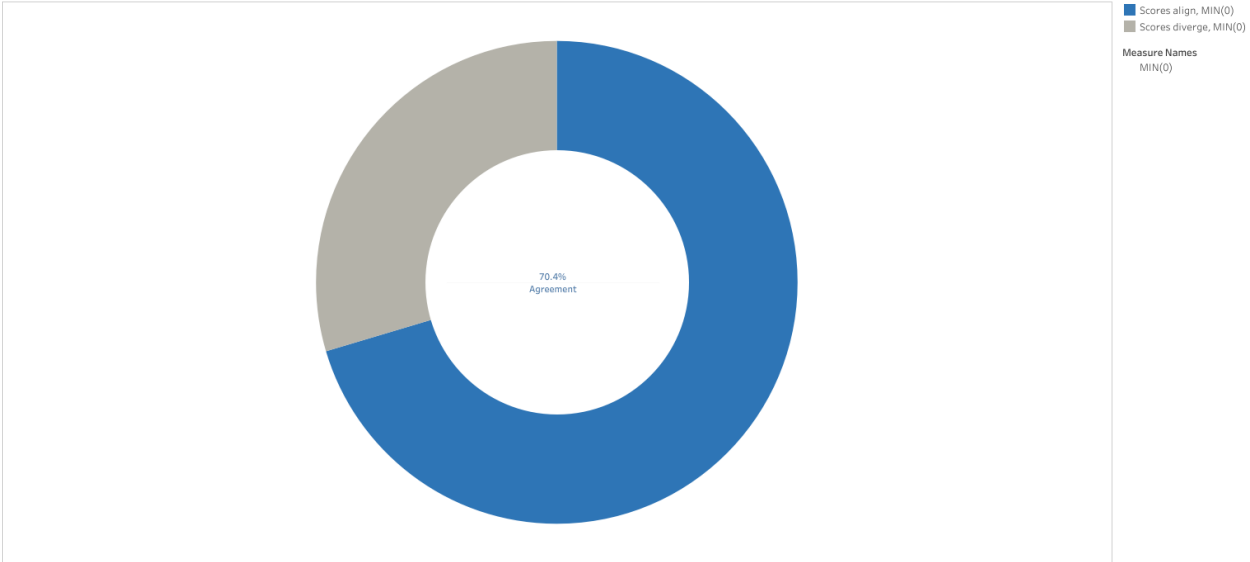
Rating	Title Count	Avg IMDB Score	% Above 7.0
TV-MA	1,814	6.63	38.0%
TV-14	1,003	6.61	41.9%
TV-PG	316	6.56	34.5%
TV-Y7	121	6.53	35.5%
TV-Y	120	6.47	33.3%
PG-13	126	6.38	24.6%
TV-G	102	6.34	27.5%
R	228	6.16	21.5%
PG	76	6.08	13.2%

Key Insight: Content maturity rating has minimal relationship with audience quality perception on Netflix. TV ratings from TV-MA to TV-Y span only 0.16 IMDB points, suggesting Netflix's TV content delivers consistent quality regardless of intended audience age. Movie ratings (R, PG-13, PG) show greater variance and all score below TV ratings, therefore reinforcing the platform-wide quality gap between TV and Movie content.

Q4.2 – IMDB vs Rotten Tomatoes Score Agreement

Q4.2 – For titles where both IMDB score and a Rotten Tomatoes score exist, how often do they agree with this being defined as both being above or below their respective averages?

Do Critics and Audiences Agree?



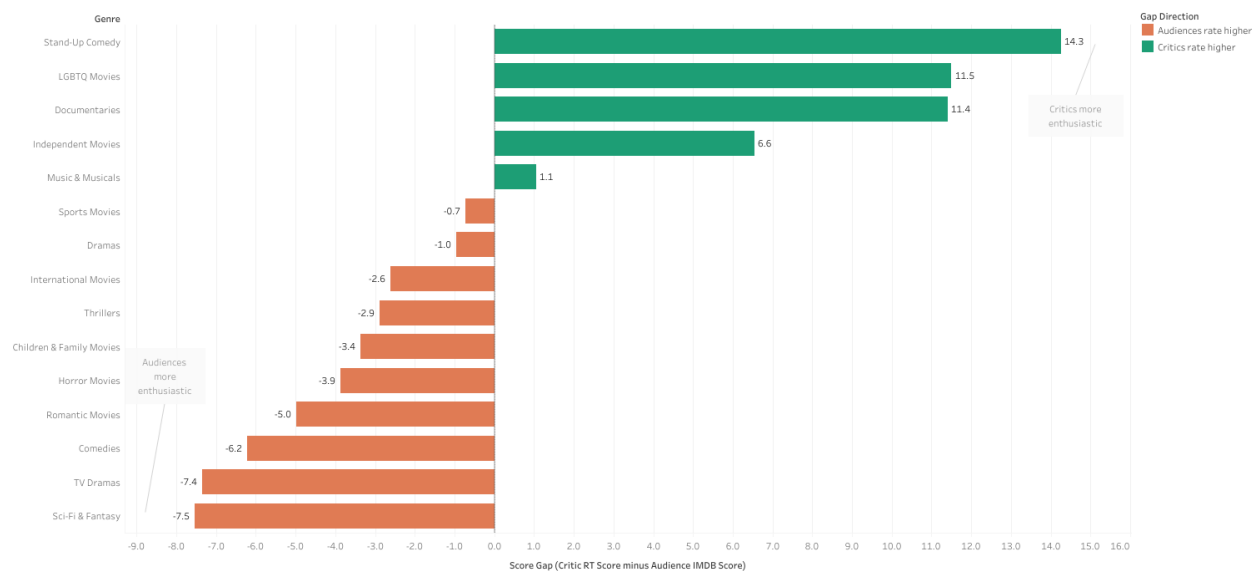
Metric	Value
Titles with both IMDB and RT scores	1,058
Scores agree (both above or below threshold)	745
Scores disagree	313
Agreement rate	70.4%
Disagreement rate	29.6%

Key Insight: Critics and audiences agree on Netflix content quality 70.4% of the time across 1,058 titled with both scores available. While substantial agreement exists, the 29.6% disagreement rate represents 313 titles where professional critics and general audiences evaluate content meaningfully different. Agreement was measured using IMDB 7.0 and RT 70 as respective quality thresholds.

Q4.3 – Critic vs Audience Gap by Genre

Q4.3 – Which genres show the biggest gap between critic scores (Rotten Tomatoes) and audience scores (IMDB), and in which direction does the gap go?

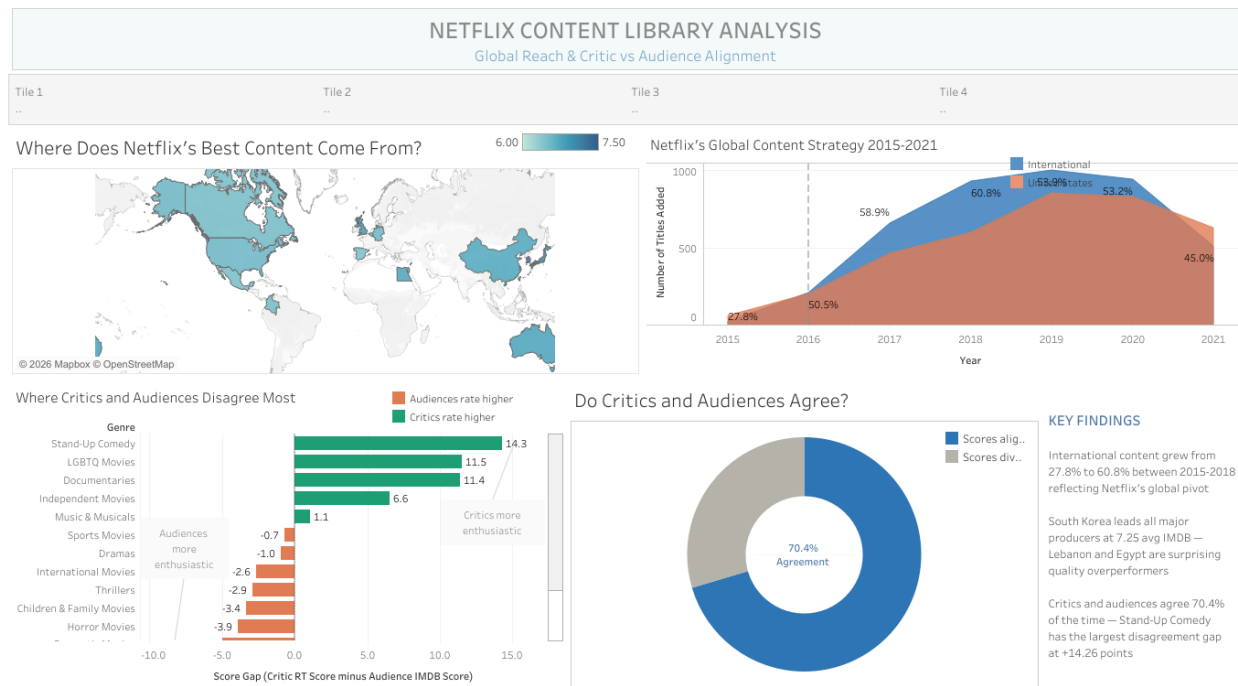
Where Critics and Audiences Disagree Most



Genre	Titles	Avg IMDB	Avg RT Score	Gap (Critics – Audience)
Stand-up Comedy	69	7.05	84.77	+14.26 (Critics higher)
LGBTQ Movies	12	6.68	78.33	+11.50 (Critics higher)
Documentaries	102	7.06	82.06	+11.41 (Critics higher)
Independent Movies	142	6.29	69.46	+6.55 (Critics higher)
Music & Musicals	60	6.42	65.23	+1.05 (Critics higher)
Dramas	477	6.47	63.73	-0.95 (Audiences higher)
International Movies	429	6.38	61.20	-2.62 (Audiences higher)
Thrillers	120	6.11	58.20	-2.88 (Audiences higher)
Horror Movies	49	5.39	50.06	-3.88 (Audiences higher)
Romantic Movies	125	6.18	56.83	-4.98 (Audiences higher)
Comedies	304	6.05	54.30	-6.21 (Audiences higher)
TV Dramas	22	7.09	63.55	-7.36 (Audiences higher)
Sci-Fi & Fantasy	52	6.10	53.42	-7.54 (Audiences higher)

Key Insight: The largest critic-audiences divergences occur at genre level. Critics significantly over-rate Stand-Up Comedy (+14.26), LGBTQ Movies (+11.50), and Documentaries (+11.41) relative to audiences. Conversely, audiences are more enthusiastic about Sci-Fi & Fantasy (-7.54), TV Dramas (-7.36), and Comedies (-6.21) than critics. Netflix faces different optimization challenges depending on whether it targets critical acclaim or audience satisfaction, two goals that are not always aligned.

Dashboard 2: Global Reach & Critic vs Audience



KEY FINDINGS:

- *International content grew from 27.8% to 60.8% between 2015-2018 reflecting Netflix's global pivot*
- *South Korea leads all major producers at 7.25 avg IMDB; Lebanon and Egypt are surprising quality overperformers*
- *Critics and audiences agree 70.4% of the time, whereas Stand-Up Comedy has the largest disagreement gap at +14.26 points*

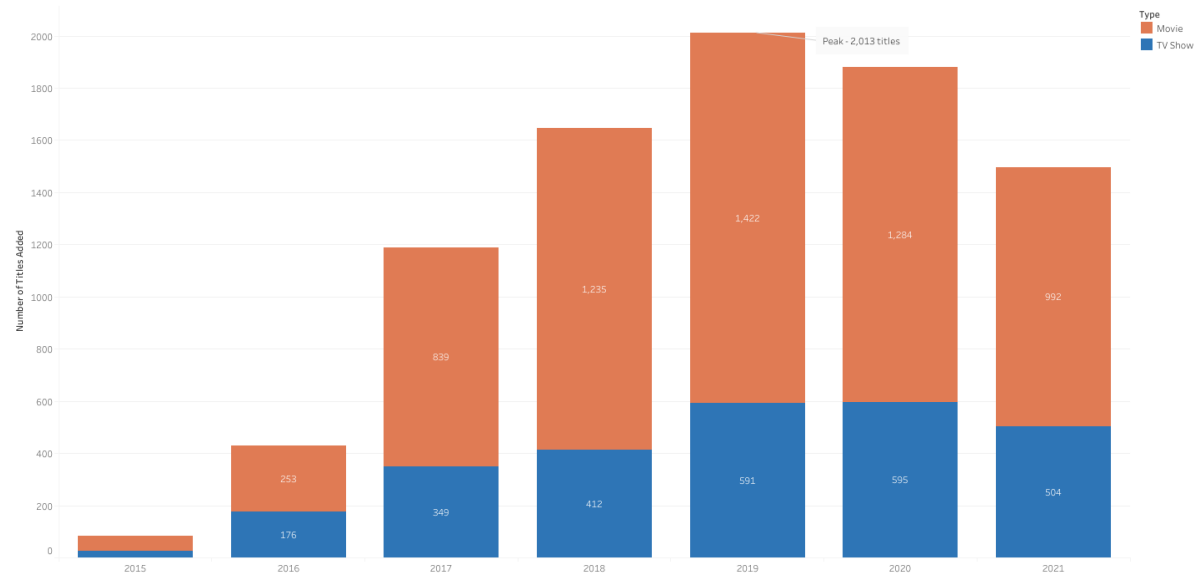
GOAL 5: LIBRARY GROWTH & QUALITY TRENDS

Charts 9-12 | Q5.1, Q5.2, Q5.3 | Dashboard 3

Q5.1 – Content Added Per Year by Type

Q5.1 – How many title were added to Netflix each year between 2015 and 2021, broken down by Movies and TV Shows, and which year had the largest single-year increase?

Netflix Content Added Per Year - Movies vs TV Shows



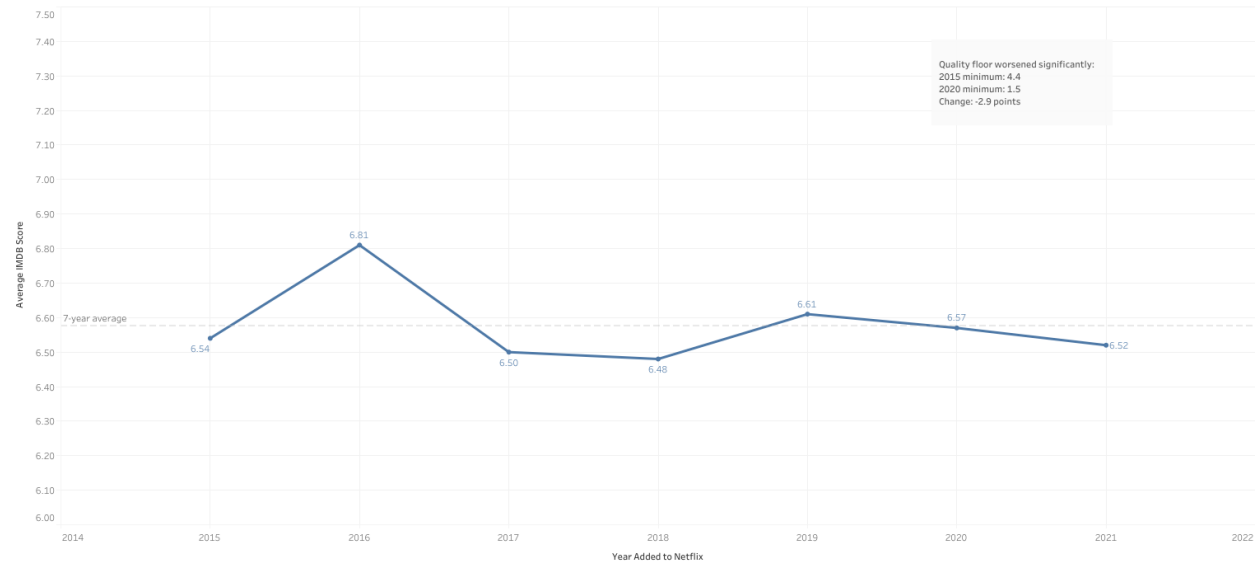
Year	Movies	TV Shows	Total	YoY Growth	TV Show %
2015	56	26	82	Baseline	31.7%
2016	253	176	429	+423%	41.0%
2017	839	349	1,188	+177%	29.4%
2018	1,235	412	1,647	+39%	25.0%
2019	1,422	591	2,013	+22%	29.4%
2020	1,284	595	1,879	-7%	31.7%
2021	992	504	1,496	-21%	33.7%

Key Insight: Netflix's content library grew explosively from 82 titles added in 2015 to a peak of 2,013 in 2019, a 2,354% increase in four years. Growth contracted in 2020 and 2021, likely reflecting COVID production shutdowns and a strategic shift toward quality over volume. TV Show additions grew from 31.7% to 33.7% of annual additions between 2015 and 2021, suggesting a gradual rebalancing toward the content format that consistently scores higher in audience quality ratings.

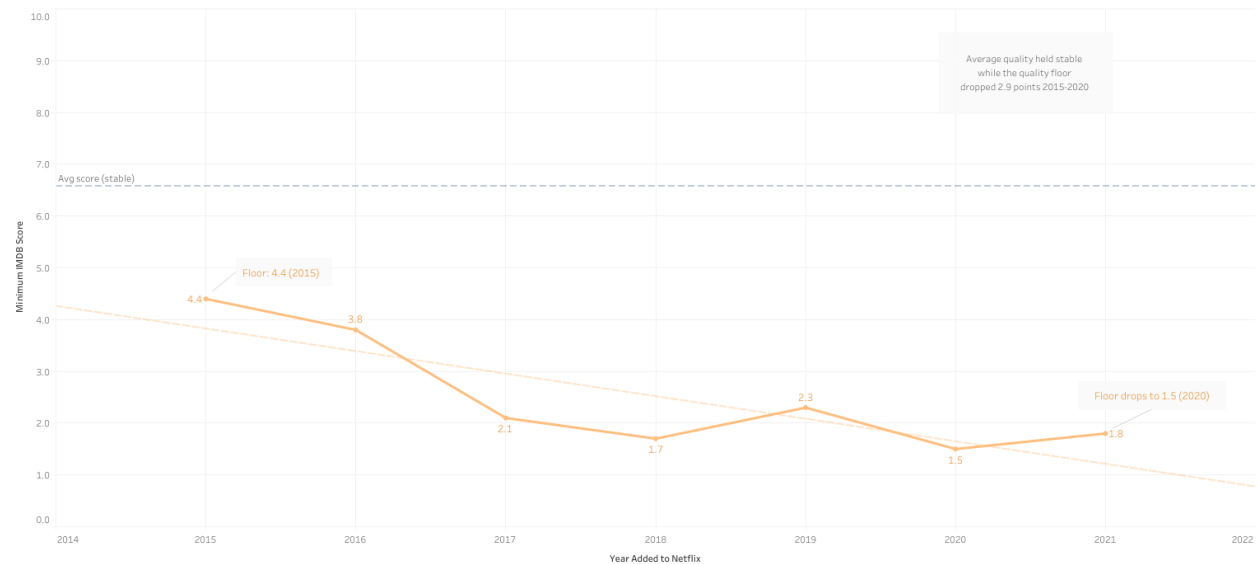
Q5.2 – Average IMDB Score Per Year Added

Q5.2 – Did the average IMDB score of Netflix content added each year increase or decrease between 2015 and 2021; and if it changed, by how much?

Netflix Quality Held Stable Despite Explosive Growth



Netflix's Quality Floor Has Deteriorated Since 2015



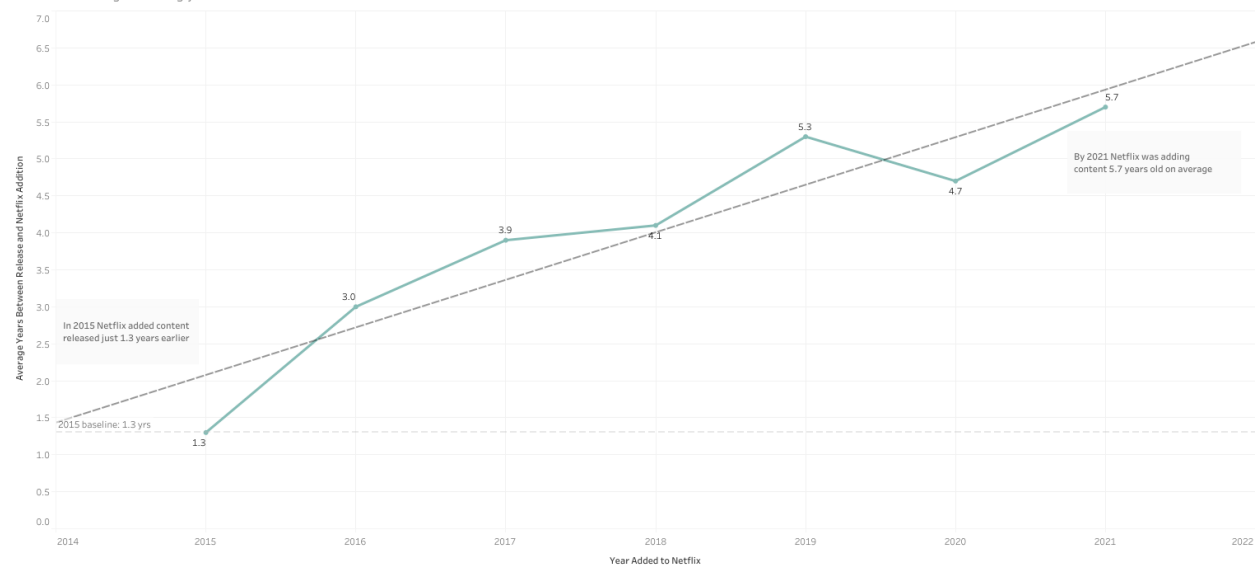
Year	Scored Titles	Avg IMDB Score	Min Score	Max Score
2015	33	6.54	4.4	8.3
2016	129	6.81	3.8	9.0
2017	358	6.50	2.1	9.0
2018	595	6.48	1.7	8.8
2019	768	6.61	2.3	9.3
2020	1,088	6.57	1.5	9.3
2021	916	6.52	1.8	9.6

Key Insight: Despite a 2,354% increase in content volume between 2015 and 2019, Netflix's average IMDB score remained remarkably stable – ranging from 6.48 to 6.81 across all years with a net change of just 0.02 points between 2015 and 2021. However, minimum scores worsened significantly from 4.4 in 2015 to 1.5 in 2020, indicating that while Netflix maintained its average quality floor, it simultaneously introduced more low-quality content at the bottom of its catalog as volume scaled.

Q5.3 – Gap Between Release Year and Netflix Addition Year

Q5.3 – What is the average gap between a title's original release year and the year it was added to Netflix, and has that gap changed over time, suggesting Netflix is licensing older or newer content?

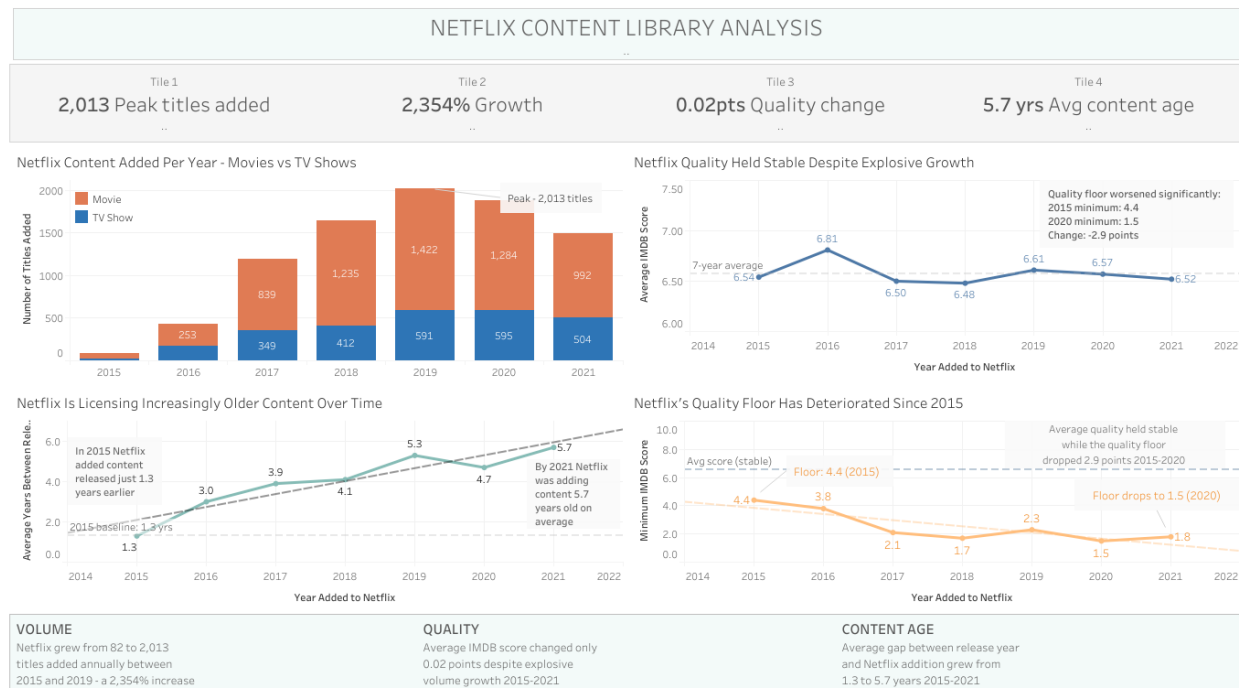
Netflix Is Licensing Increasingly Older Content Over Time



Year Added	Avg Gap (Years)	Min Gap	Max Gap	Title Count
2015	1.3	0	23	82
2016	3.0	0	43	427
2017	3.9	0	75	1,187
2018	4.1	0	93	1,644
2019	5.3	0	64	2,009
2020	4.7	0	66	1,876
2021	5.7	0	76	1,496

Key Insight: The average gap between a title's original release year and its Netflix addition year grew from 1.3 years in 2015 to 5.7 years in 2021 which suggest that Netflix is increasingly licensing older back catalog content as it scaled its library. This strategy likely contributed to quality stability despite volume growth, as older content has already demonstrated audience appeal over time. Maximum gaps exceeding 75 years in every year from 2017 onwards confirm Netflix actively acquires archival and classic content alongside contemporary productions.

Dashboard 3: Library Growth & Quality Trends



KEY FINDINGS:

- **VOLUME** – Netflix grew from 82 to 2,013 titles added annually between 2015 and 2019: a 2,354% increase
- **QUALITY** – Average IMDB score changed only 0.02 points despite explosive volume growth 2015-2021
- **CONTENT AGE** – Average gap between release year and Netflix addition grew from 1.3 to 5.7 years 2015-2021

3. Consolidated Findings & Strategic Recommendations

3.1 Key Findings Summary

Goal	SMART Questions	Primary Findings
Goal 1 – Content Quality	Q1.1, Q1.2, Q1.3	TV Shows significantly outperform Movies (7.05 vs 6.26). Top 10 titles exclusively TV Shows. 74.8% of scored content has under 10k votes.
Goal 2 – Genre Strategy	Q2.1, Q2.2	Most produced genres are not highest rated Korean TV Shows lead at 7.47 with only 72 titles. Every top-15 rated genre is TV format.
Goal 3 – Global Reach	Q3.1, Q3.2, Q3.3	International content grew from 27.8% to 60.8% between 2015-2018. South Korea and Japan lead quality. Lebanon and Egypt are surprising overperformers.
Goal 4 – Critic vs Audience	Q4.1, Q4.2, Q4.3	70.4% agreement between critics and audiences. Content rating barely affects quality. Stand-Up Comedy has largest critic-audience gap at +14.26.
Goal 5 – Trends Over Time	Q5.1, Q5.2, Q5.3	2,354% volume growth with only 0.02 point quality decline. Minimum scores worsened. Average content age gap grew from 1.3 to 5.7 years.

3.2 Strategic Recommendations

Based on the data analysis, the following strategic recommendations can be made to Netflix regarding content investment:

- Increase Korean TV Show investment – highest rated genre at 7.47 avg with only 72 titles. Significant whitespace exists relative to quality demonstrated.
- Reduce low-quality Movie volume – International Movies and Comedies produce highest title counts with below-average scores. Quality-focused curation over volume would improve platform perception.
- Expand TV Thriller and Anime catalogs – both genres show strong audience reception (7.39 and 7.28) with limited content supply
- Leverage Middle Eastern content – Lebanon and Egypt outperform much larger markets in quality scores with minimal current investment. Untapped regional potential exists.
- Maintain TV Show growth trajectory – the gradual shift toward TV content (31.7% to 33.7%) aligns with the consistent quality advantage TV Shows hold over Movies. Accelerating this shift could improve overall platform ratings.

- Address quality floor deterioration – minimum IMDB scores dropped from 4.4 in 2015 to 1.5 in 2020, indicating increasingly poor bottom-end content. A content quality threshold policy could prevent further deterioration.

3.3 Data Limitations

- IMDB scores are only available for 47.3% of titles – analysis involving scores may be more biased toward more widely known content.
- Rotten Tomatoes scores are available for only 25.8% of titles – critic vs audience comparison is limited to a subset of the library.
- The dataset covers content through approximately mid-2021 – 2021 figures are partial year and show not be compared directly to full prior years.
- Title matching across datasets was performed on title text – minor title variations may have resulted in some titles not matching correctly across sources.
- Country field represents production country, not audience country – conclusions about regional performance reflect content origin not viewership geography.

4. Interactive Dashboard

The complete interactive dashboard for this analysis is publicly available on Tableau Public. The dashboard consists of three pages navigable via the tab bar at the bottom of the screen.

Dashboard	Charts	Goals Covered
Dashboard 1 – Quality & Genre	Charts 1-4	Goal 1 – Content Quality Assessment, Goal 2 – Content Strategy by Genre
Dashboard 2 – Global & Critics	Charts 5-8	Goal 3 – Global Content Reach, Goal 4 – Critic vs Audience Alignment
Dashboard 3 – Trends Over Time	Charts 9-12	Goal 5 – Library Growth & Quality Trends Over Time

Tableau Public URL:

<https://public.tableau.com/app/profile/geri.rusk/viz/NetflixContentLibraryAnalysis-GeriRusk/QualityGenre#1>

5. Chart & Screenshot Reference

The following table maps every screenshot included in this report to its corresponding SMART question, business goal, and source.

Screenshot Title	SMART Question	Source
Q1.1 – TV Shows vs Movies Average IMDB Score	Q1.1	Chart 1 – Tableau Dashboard 1
Q1.2 – Top 10 Highest Rated Titles on Netflix	Q1.2	BigQuery result – Goal 1 SQL
Q1.2 – Bottom 10 Lowest Rated Titles on Netflix	Q1.2	BigQuery result – Goal 1 SQL
Q1.3 – Content Popularity vs Average IMDB Score	Q1.3	Chart 2 – Tableau Dashboard 1
Q2.1 – Genre Volume vs Average IMDB Score	Q2.1	Chart 3 – Tableau Dashboard 1
Q2.2 – Highest Rated Underproduced Genres	Q2.2	Chart 4 – Tableau Dashboard 1
Q3.1 & Q3.3 – Content Quality by Country	Q3.1, Q3.3	Chart 5 – Tableau Dashboard 2
Q3.2 – International vs US Content Growth	Q3.2	Chart 6 – Tableau Dashboard 2
Q4.1 – Average IMDB Score by Content Rating	Q4.1	BigQuery result – Goal 4 SQL
Q4.2 – Critic vs Audience Agreement Rate	Q4.2	Chart 8 – Tableau Dashboard 2
Q4.3 – Critic vs Audience Score Gap by Genre	Q4.3	Chart 7 – Tableau Dashboard 2
Q5.1 – Netflix Titles Added Per Year by Type	Q5.1	Chart 9 – Tableau Dashboard 3
Q5.2 – Average IMDB Score Stability 2015-2021	Q5.2	Chart 10 – Tableau Dashboard 3
Q5.2 Supplementary – Quality Floor Deterioration	Q5.2	Chart 12 – Tableau Dashboard 3
Q5.3 – Content Age Gap Over Time	Q5.3	Chart 11 – Tableau Dashboard 3